

CIS 635

Knowledge Discovery & Data Mining

Fall 2026 Syllabus

Updated: August 26, 2026

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Meeting Times: Section 01: T 6:00 pm–8:50 pm; Section 02 MW 4:30 pm–5:45 pm (DCIH 420)
Office Hours: Online / MAK D-2-216
([schedule by appointment](#))

1. General Information

1.1 Course Objectives

After completing this course, students should be able to:

- Wrangle, analyze, and visualize data, primarily using Python.
- Exercise and apply supervised and unsupervised knowledge discovery techniques.
- Identify and build models using common data mining techniques.
- Evaluate models for their effectiveness and appropriateness.
- Communicate findings using effective visualizations.
- Critically evaluate knowledge-discovery methods and communicate defensible conclusions based on data, model performance, assumptions, and limitations.

1.2 Course Description

This course provides a survey of computational methods for knowledge discovery and data mining, with an emphasis on applications in bioinformatics, medicine, and other data-intensive domains. Students develop practical skills in Python-based data analysis, data preparation and visualization, predictive modeling, model evaluation, clustering, dimensionality reduction, sequence analysis, neural networks, and association analysis. The course also introduces domain-specific computational methods, including dynamic programming for sequence alignment and similarity search (e.g., BLAST), hidden Markov models for sequence and structure analysis, and clustering and classification methods for biological and gene-expression data. Students gain hands-on experience working with biological databases and selected data-mining software while completing end-to-end knowledge discovery tasks involving real-world datasets.

1.3 Requirements and Prerequisite

Students need an internet connection, a computer, access to Blackboard, and the ability to run Python and install required Python packages. If you have difficulty accessing these resources or meeting these requirements, contact the instructor so that an appropriate solution can be explored.

Prerequisite: N/A

1.4 Delivery Format

The course is delivered primarily through in-person class meetings, supported by Blackboard for course materials, assignments, announcements, and grades.

1.5 Recommended References

Specific references will be shared throughout the course. The following resources are recommended:

- *Data Mining: A Knowledge Discovery Approach* by Witold Pedrycz and collaborators (available online through the GVSU Library).
- *Introduction to Data Mining*, 2nd Edition, by Pang-Ning Tan, Michael Steinbach, and Vipin Kumar.
- *Pattern Recognition and Machine Learning* by Christopher M. Bishop.
- *Machine Learning with Python Cookbook* by Chris Albon.
- *Fundamentals of Data Visualization* by Claus O. Wilke.
- *Python for Data Analysis* by Wes McKinney.
- *Automate the Boring Stuff with Python* by Al Sweigart.
- *Python Data Visualization Essentials Guide* by Kalilur Rahman.

2. Tentative Schedule

Subject to change.

Week	Week of	Topics
1	Aug. 31	Introduction to Knowledge Discovery & Data Mining; Python Basics
2	Sept. 7	Python for Data Science; Descriptive Statistics & Data Visualization
3	Sept. 14	Exploratory Data Analysis (EDA) I
4	Sept. 21	Exploratory Data Analysis (EDA) II & Case Studies
5	Sept. 28	Similarity & Distance Measures; K-Nearest Neighbors
6	Oct. 5	Predictive Modeling: Regression
7	Oct. 12	Predictive Modeling: Classification I
8	Oct. 19	Predictive Modeling: Classification II & Model Evaluation
9	Oct. 26	<i>Fall Break</i>

Week	Week of	Topics
10	Nov. 2	Unsupervised Learning: Clustering
11	Nov. 9	Clustering & Dimensionality Reduction
12	Nov. 16	Sequence Analysis & Modeling
13	Nov. 23	Special Topics: Deep Learning & Neural Networks
14	Nov. 30	Special Topics: Emerging Data-Mining Techniques
15	Dec. 7	Final Project Presentations & Course Wrap-Up
16	Dec. 14	Final Examination & Evaluation Week

Final Examination Period

The CIS 635 final examination will be administered during the University examination period. Class instruction ends December 12, 2026, and the examination period is December 12–19, 2026. The official final-exam date, time, and location will follow the University final-examination schedule and will be communicated through Blackboard.

3. Course Organization and Grading Format

3.1 Grading Distribution

The final course grade will be determined using the following grading distribution:

Assessment	Weight
Testing exercises / programming assignments	30%
Exams / quizzes	40%
Team KDD project	20%
Final presentation / advanced analysis	10%
Total	100%

3.1.1 Homework Assignments and Tests

Assignments and/or class tests will reinforce course material. They will typically include a mixture of programming problems and mathematical or written questions. Unless otherwise stated, individual assignments must be completed independently. Students may discuss higher-level concepts and ask questions, but may not share code or responses or review another student's code or responses.

3.1.2 Final Project

Students will work in groups on a knowledge discovery and data mining project. Each group will identify a problem and associated dataset(s) related to a domain of expertise and/or interest and complete an end-to-end KDD project. The project emphasizes a learning-by-doing approach.

Each team will deliver a final presentation and submit project documentation. All members must explicitly describe their individual and collective contributions to the project.

3.2 Grading Scale

For final grade calculation, any fractional final percentage will be rounded up to the next whole percentage. For example, 89.1% will be recorded as 90%.

Grade	Minimum	Grade	Minimum
A	93%	C	73%
A-	90%	C-	70%
B+	87%	D+	67%
B	83%	D	60%
B-	80%	F	Below 60%
C+	77%		

4. Course Expectations, Policies, and Academic Honesty

4.1 Expectations

To support your success in this course, you are expected to:

- Check Blackboard regularly for announcements, course materials, assignments, and grades.
- Stay up to date with required course videos and readings.
- Communicate feedback about the course and teaching that may help improve the learning experience.
- Adhere to CIS and GVSU policies on academic honesty.

4.2 Course Policies

- Weekly reflections and homework assignments must be completed individually unless otherwise stated.
- **Due dates:** Specific due dates will be posted with each assignment in Blackboard. Unless otherwise announced, assignments are due at 11:59 p.m. Eastern Time on the stated due date.
- **Late policy:** 10% of the maximum possible grade is deducted per day late, up to three days (30% total). After three days, the assignment will not be accepted.
- The instructor reserves the right to modify course policies, the course calendar, and due dates as needed.
- This course is subject to the GVSU course policies listed at www.gvsu.edu/coursepolicies/.

4.3 Use of Generative AI Tools

Generative AI tools may be used only as permitted for individual course activities and assignments. Students remain responsible for understanding, validating, and being able to explain all submitted work, including code, analyses, visualizations, and written material. AI-generated content or code must not be submitted as the student's own work when prohibited by assignment instructions. When AI assistance is used in a permitted manner, students may be required to disclose the tool

used and the nature of the assistance. Specific AI-use expectations will be provided with individual assignments and assessments.

4.4 Academic Honesty

All students are expected to adhere to the academic honesty standards of Grand Valley State University and the academic honesty guidelines of the School of Computing. Details are available at <https://www.gvsu.edu/computing/academic-honesty-30.htm>.

Collaboration and learning from peers and the broader community are encouraged, but presenting another person's work as your own is prohibited. For individual assignments:

- Document collaboration.
- Do not electronically transfer code between students.
- Cite code found on the internet with an active link to the source. External code must not solve the entirety of an assigned problem or project.
- You may participate in online forums, but do not post solutions or solicit others to complete your work.
- You may discuss problems with peers in non-technical terms, but you remain responsible for your own submission.
- You are ultimately responsible for all aspects of your submissions and should be able to explain and defend your work.

5. Useful Resources

5.1 Blackboard

Course materials, assignments, grades, and announcements will be posted to Blackboard (<https://lms.gvsu.edu/>). Students are responsible for staying informed. Blackboard announcements are used to communicate timely information, including changes to deadlines; students should ensure their notification settings allow them to receive updates.

5.2 Other Academic Resources

GVSU provides academic-support resources, including:

- The Writing Center
- Speech Lab
- Research Consultants
- Library Liaisons

5.3 Disability Support

Students who need accommodations should present a memo from Disability Support Resources (DSR) describing the approved reasonable accommodations. If you have not already done so, con-

tact Disability Support Resources at 215 CON, by phone at 331-2490, or by email at dsrgvsu@gvsu.edu. The instructor cannot provide disability accommodations until a DSR-issued memo has been received. All discussions will remain confidential. More information is available at <https://www.gvsu.edu/dsr/>.

5.4 Religious Observance

Many University students, staff, and faculty observe religious traditions. The Religious Inclusion Policy acknowledges the right of students, staff, and faculty to engage in religious observances. If you need special accommodations to observe a religious holiday, please make arrangements with the instructor in advance.

5.5 In Case of Emergency

Fire: Immediately proceed to the nearest exit when a fire alarm sounds. Do not use elevators. Additional information is available on the University Emergency website at gvsu.edu/emergency.